

# ValidateHE Validation Report — METIMMOX-1

## Transparency & verification of a biomarker-guided immunotherapy cost-effectiveness model (MSS/pMMR mCRC)

ValidateHE v1 (AI-assisted validation; reviewed model commit e098658)

2026-07-12

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### 1. Executive summary (non-technical)

**Model:** METIMMOX-1 — partitioned survival model (PSM) model at `c:/Users/benjampg/git/METIMMOX-1/mod` (commit e098658, `model/` is an untracked subdirectory of the parent repo).

**Requested modules:** Module 1 (Transparency) and Module 3 (Verification). Modules 2 (Face validity), 4 (Cross-validation), and 5 (External validation) were not requested.

**Overall verdict: Fail**

The model is internally **not verified**: its core partitioned-survival partition is built from overall-survival (OS) and progression-free-survival (PFS) curves that are fit as two independent parametric models with no ordering constraint, and for the two biomarker-positive economic subgroups (CRP+, TMB/BRAF+) the fitted PFS curve exceeds the OS curve across roughly half of the 10-year extrapolated horizon. A silent `pmin(pfs, os)` clamp, surfaced only as an easily-missed `warning()`, forces the “progressed” health state to zero occupancy for that entire span — distorting the quality-adjusted life-year (QALY) and cost partition between progression-free and progressed care for exactly the arms the economic strategies are built to evaluate. Separately, the supporting documentation is internally inconsistent: a June 2026 single-model/no-TLR refactor was applied to the code and to `AGENTS.md` and the currently-rendered reports, but never propagated to the sibling `docs/README.md` or the in-repo manuscript, which still describe a three-biomarker, Model-A/B/C structure that no longer exists in code; and the documented default utility source (IPD-derived) contradicts the code’s actual default (CORRECT trial), blocking reproduction of the manuscript’s headline QALYs/ICERs from a fresh clone with default settings. Both defects were confirmed by independent evidence streams (black-box behavioural testing plus white-box code inspection for the curve-crossing issue; two independent documentation reviewers for the contradictions) and neither is disclosed as a known limitation anywhere in the reviewed materials. No file in the reviewed repository was modified by this review (integrity hash confirmed unchanged before and after).

## What was and wasn’t assessed

**Assessed (this run):** Module 1 (transparency/documentation review) was performed against 20 auto-discovered and in-repo documentation sources (`AGENTS.md`, sibling `docs/README.md`, `CLAUDE.md`, `LICENSE`, the in-repo Lancet Oncology manuscript draft, 12 rendered Quarto report families, four technical-documentation reports, and source/roxygen comments), by two independent LLM reviewers (documentation-completeness and documentation-consistency). Module 3 (verification) was performed on a read-only scratch copy of the model: behavioural black-box testing (56 tests including a base-case run, trace/partition invariants, extreme-value tests, a PSA batch of 100 iterations, and 9 unit-style tests of core functions), white-box code inspection (12 findings, including root-causing the black-box failures), and a coding-standards/CHAOS review (14 findings). The confidential patient-level trial data (`data/tidy/METIMMOX.rds`, `data/sensitive/*.xlsx`) was present locally, so the model could be executed rather than only statically inspected.

**Not assessed:** Module 2 (AI-simulated face-validity council), Module 4 (cross-validation against a comparator model), and Module 5 (external validation against independent observed event data) were **not requested** for this run; every checklist item owned by those modules is marked **Deferred** below rather than scored. No comparator model or external validation dataset was supplied to this run, and none was auto-discovered as reviewable evidence in the repository beyond a single narrative sentence in the manuscript comparing predicted 5-year OS to a published registry figure (which Module 1 records but does not itself validate). “Fable\_5” in this report’s naming refers to the AI model that performed this ValidateHE run (Claude Fable 5), not to a model-validation grading scale.

**Transparency disclosure:** This module is an AI-assisted documentation-completeness and documentation/code-consistency review. It establishes whether documentation is complete, findable, and consistent with static code artifacts — it does **not** by itself establish clinical validity or correct implementation (see Module 3 for that).

**Face-validity disclosure:** Not requested for this run.

**Cross-validation disclosure:** Not requested for this run.

**External-validation disclosure:** Not requested for this run.

No file in the reviewed repository was modified: all testing ran on a temporary copy, and a before/after content hash of the original confirmed it is byte-for-byte unchanged. All temporary artifacts are deleted as the final step of the run; if the run instead ends with an integrity warning, the scratch copy is retained and its path printed. **No fixes were applied** — recommended fixes are listed in Appendix A only.

### Verdict by method

| Method                          | Verdict                               | Tests run | Pass | Partial | Fail | Not-tested |
|---------------------------------|---------------------------------------|-----------|------|---------|------|------------|
| Transparency document review    | Insufficiently transparent for review | 33        | 7    | 14      | 11   | 0          |
| Face-validity council           | Not requested (Deferred M2)           | 0         | 0    | 0       | 0    | 0          |
| Black-box (behavioural)         | Fail                                  | 56        | 43   | 0       | 10   | 3          |
| White-box (code inspection)     | Fail                                  | 12        | 5    | 4       | 3    | 0          |
| Coding standards (darth)        | Adequate with reservations            | 14        | 2    | 9       | 1    | 1          |
| Cross-validation (comparative)  | Not requested (Deferred M4)           | 0         | 0    | 0       | 0    | 0          |
| External validation (empirical) | Not requested (Deferred M5)           | 0         | 0    | 0       | 0    | 0          |

*(Transparency: 1 additional finding scored NA, not shown in the 4 count columns. Coding standards: 1 additional finding scored NA, not shown in the 4 count columns.)*

## Checklist rollup

| Checklist                     | Pass | Partial | Fail | NA | Not-tested | Deferred |
|-------------------------------|------|---------|------|----|------------|----------|
| CDA-AMC Model Validation Tool | 11   | 9       | 14   | 4  | 17         | 2        |
| TECH-VER                      | 27   | 8       | 7    | 0  | 5          | 3        |
| AdViSHE                       | 2    | 7       | 5    | 0  | 0          | 1        |
| CHAOS v0.9                    | 3    | 5       | 2    | 0  | 0          | 0        |

*Deferred = owned by a module that was not requested for this run. NA = item cannot apply to an R model (e.g., Excel-specific); R-equivalents, where they exist, were scored instead — see [references/checklist-routing.md](#).*

## Top findings in plain language

- **We checked whether the progression-free-survival (PFS) curve ever exceeds the overall-survival (OS) curve** (a health-economic model should never have “more time free of progression” than “time alive”). It should not, by definition. Instead, for the CRP-guided and TMB/BRAF-guided biomarker-positive subgroups — the two arms the entire economic evaluation exists to assess — PFS exceeds OS across roughly **49-51% of the 10-year modeled horizon**. Root cause: OS and PFS are fitted as two separate parametric survival models, each independently chosen by AIC (`06_parametric_survival_analysis.R`), with nothing constraining the extrapolated curves to preserve OS > PFS (findings `BB-PSM-ORD-crp`, `BB-PSM-ORD-tmb_braf`, `WB-001`). This is the single most important finding in this review and is why Module 3 verdicts Fail.
- **We checked how the code handles that curve crossing**. It applies `pfs <- pmin(pfs, os)` and emits a bare `warning()` — a message easily lost in a batch run or a 5,000-iteration PSA — which silently forces the “progressed” health state to exactly zero occupancy for the entire crossing region rather than surfacing the magnitude of the correction (`WB-003`). Because post-progression utility and end-of-life/follow-up costs differ materially from progression-free values, this silently redistributes QALYs and costs for the biomarker-positive arms without any reported flag of how much of the horizon was affected.
- **We checked whether documentation matches the deployed code for the set of biomarker strategies evaluated**. The currently-rendered reports and `AGENTS.md` consistently describe two economic strategies (CRP, TMB/BRAF) with TLR excluded by design. `docs/README.md` and the in-repo Lancet Oncology manuscript draft, however, still describe a third TLR-guided strategy and a “Model A/B/C” multi-structure comparison that a June 2026 refactor removed from the code (`TR-CONS-001`, `TR-CONS-002`). The manuscript’s Model A/C and TLR-strategy headline results are therefore **not reproducible from the current repository under any settings**.
- **We checked which utility values the default configuration actually produces**. `AGENTS.md`’s parameter table documents the default as IPD-derived utilities (`u_np=0.9077`, `u_p=0.9005`), but the code’s actual default (`UTILITY_SOURCE <- 1`) uses CORRECT-trial values (`u_np=0.73`, `u_p=0.59`); the manuscript’s reported utilities (0.91/0.90) match the *documented* default, not the *code*’s default (`TR-CONS-003`). A fresh clone run with default settings will not reproduce the manuscript’s headline QALYs/ICERs.

- **We checked whether swapping two strategies’ clinical-effect inputs produces a clean swap of outputs** (a standard verification test). QALYs and life-years swapped exactly, confirming the survival-curve/prevalence logic is sound, but costs did not: the CRP and TMB/BRAF diagnostic-test costs are attached by matching a hard-coded biomarker **name string** in `calculate_outcomes.R`, not by data carried with the strategy (BB-SW, WB-004). Not a base-case defect today, but a latent coupling that would silently misattribute costs if strategies are ever renamed or reordered.
- **We checked reproducibility infrastructure.** No `renv.lock`, no `DESCRIPTION` file, and no in-repo `README.md` (present in git history but currently deleted from the working tree) exist anywhere in the reviewed copy; all 37 detected packages are loaded via `pacman::p_load()` with no pinned versions (CS-005). Combined with the confidential trial data being intentionally excluded from the public repository, full independent reproduction by an outside party is not currently possible even though the documented run order and code structure are otherwise clear.
- **We checked whether any documented, independent human expert review of the conceptual model, input data, or implemented code exists.** None was found (TR-COMP-011, TR-COMP-013, TR-COMP-015 — AdViSHE A1/B1/C1); the manuscript’s own “Contributors” section is a [To be completed] placeholder.
- **On the positive side**, a 100-iteration PSA batch completed cleanly (0 run errors, 0% fallback-to-base-case-curve rate), the base case is fully deterministic on re-run, cohort conservation and state-occupancy bounds held for every strategy and every cycle, and CDA-AMC PSA-consistency items 43-46 all passed — the model’s plumbing (discounting, cohort accounting, PSA sampling machinery) is sound; the defect is specifically in how the OS/PFS curves are extrapolated and reconciled.

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## 2. Technical body

### 2.1 Environment manifest

- **Generated:** 2026-07-12T02:57:54+0200 (Europe/Oslo)
- **R:** 4.3.2 on x86\_64-w64-mingw32
- **Quarto:** 1.6.40
- **Original repo:** C:/Users/benjampg/git/METIMMOX-1/model
- **Scratch copy:** C:/Users/benjampg/AppData/Local/Temp/claude/c--Users-benjampg-git-METIMMOX-1

### Git (read-only capture of the original repo)

- not under version control (path sits inside unrelated repo C:/Users/benjampg/git/METIMMOX-1@e098658419)

### Detected tech stack

- `flexsurv/flexsurvreg` -> parametric survival extrapolation
- `survival` -> time-to-event analysis

- dampack -> CEA post-processing (ICERs, CEAC, PSA)
- Quarto/Rmd -> literate reporting

**Model-type hint:** markov (heuristic; confirmed by review to be a partitioned survival model)

## Packages

DESCRIPTION present: FALSE; `renv.lock` present: FALSE. All 37 detected packages are flagged **undeclared** (no version pinning anywhere): coin, cowplot, coxphf, dagitty, dampack, dathtools, devtools, dplyr, emmeans, flexsurv, gems, ggdag, ggplot2, ggtext, glmnet, grid, gridExtra, here, kableExtra, knitr, logistf, MASS, Matrix, mstate, mvtnorm, pacman, readr, readxl, reshape2, scales, survival, survminer, tableone, tibble, tidyr, tidyverse, vcd, xtable. (One additional row, `pkg`, is a false positive from the inventory tool’s static parser — a for-loop variable name in `snapshot_utils.R:31`, not a real dependency; see CS-014.)

## Folder tree (261 entries)

The reviewed copy follows a DARTH-style layout: `data/{output,raw,sensitive,tidy}`, `figs/`, `manuscript/`, `references/`, `scripts/QMD/{report,technical_docs,vignettes}`, `scripts/R/{analysis,archive,functions,tests}`, `tables/`. Numbered pipeline scripts run `01_data_prep.R` through `15_save_snapshot.R`; reusable functions live in `scripts/R/functions/`; deprecated code is isolated in `scripts/R/archive/` (confirmed unreferenced). See Appendix B for the full function inventory and dependency graph.

## 2.2 Method: how this review was performed

This run used 5 isolated review agents plus orchestrator-performed inspection: two independent Module-1 (transparency) documentation reviewers (Sonnet) — one scoring documentation *completeness*, one scoring documentation/code *consistency*; and three Module-3 (verification) subagents — black-box behavioural testing (Sonnet), coding-standards/CHAOS review (Sonnet), and white-box code inspection. The white-box subagent was dispatched but was interrupted by a session limit before it could write findings; the white-box results in this report were instead produced by the ValidateHE orchestrator (Claude Fable) performing the deep root-cause code inspection directly, cross-referencing the black-box subagent’s failures against the source code line-by-line. This substitution is disclosed here transparently rather than silently backfilled. All review agents worked exclusively on a read-only scratch copy; no file in the original repository was read-write accessed or modified.

### Module 1: transparency

Two LLM reviewers independently read the documentation corpus listed in §2.1/Appendix, without access to each other’s output, then their findings were reconciled by the orchestrator. The completeness reviewer scored whether each CDA-AMC/AdViSHE/CHAOS documentation-layer item is present and findable; the consistency reviewer specifically hunted for contradictions between documents and between documentation and static code (e.g., grepping `model_configs.R` against

README claims, comparing git-blame timestamps against documented defaults). Both reviewers independently converged on the same core problem before being reconciled, which is treated as corroborating evidence rather than double-counting.

## **Module 2: face validity**

Not requested.

## **Module 3: verification**

Black-box testing ran the model's base case and 27 additional behavioural probes (equalized effectiveness, extreme utilities/costs/mortality, discount-rate extremes, horizon changes, arm-swap, doubling/halving of effect size, a 100-iteration PSA batch) plus 9 unit-style tests of `validate_model_params()`, `calculate_outcomes()`, and `extract_all_survival_probabilities()`, all executed against the actual R code on the scratch copy (not simulated). White-box inspection traced the root cause of every black-box failure back to specific file:line locations in `model_fun.R`, `calculate_outcomes.R`, and `06_parametric_survival_analysis.R`, and independently verified the discounting formula, partition-sums-to-1 identity, and forward/backward traceability of two representative parameters. Coding-standards review scored the CHAOS v0.9 rubrics (folder structure, style, documentation, modularity, reproducibility) plus TECH-VER Stage-5 programming/efficiency concerns (DRY violations, loop idioms, error-masking patterns, hidden global dependencies).

## **Module 4: cross-validation**

Not requested.

## **Module 5: external validation**

Not requested.

## **2.3 Transparency results**

**Documents reviewed (20 source ids):** 4 auto-discovered sibling docs (`AGENTS.md`, `docs/README.md`, `CLAUDE.md`, `LICENSE`), the in-repo manuscript (Lancet Oncology v4 draft), 12 Quarto report families (plus their rendered PDFs, extracted to text via `pdftotext`), 4 technical-documentation reports (`bug_fix_impact`, `age_effect_analysis`, `all_parametric_survival_models`, `model_b_comparison`), the bibliography, and roxygen/source comments. No `--docs` argument was supplied; sibling documentation was auto-discovered from the parent case directory (`docs/` sits alongside `model/`). Absent from the corpus: `DESCRIPTION`, `renv.lock`, `NEWS/CHANGELOG`, an in-repo `README.md` (it lives only in the sibling `docs/` folder), and a `METIMMOX.rds` data dictionary.

**Consensus (both reviewers independently converged):** a June 2026 single-model/no-TLR refactor (parent-repo commit `ab7a079`, "refactor stage 7: finish single-model/no-TLR cleanup, fix

renders, regen outputs”) was propagated to `AGENTS.md` and the currently-rendered economic reports, but **not** to the sibling `docs/README.md` or the in-repo manuscript, leaving three mutually inconsistent vintages of the strategy set describing the model. Five candidate contradictions were independently **confirmed** with file:line and git-history evidence:

1. `docs/README.md` lists TLR as a third economic strategy versus `model_configs.R:28` `ALL_STRATEGIES <- c("control", "crp", "tmb_braf")` (TR-CONS-001).
2. `docs/README.md` and the manuscript describe a “Model A/B/C” multi-structure comparison that no longer exists in code — `multi_model_cea.R` (despite its name) contains only single-model helper functions per its own header comment; the manuscript’s Model A/C and TLR-strategy headline results are structurally unreproducible from current code (TR-CONS-002, TR-CONS-007).
3. `AGENTS.md`’s documented `UTILITY_SOURCE` default (0, IPD-derived) contradicts the code’s actual default (1, CORRECT trial) at `02_setup_and_global_variables.R:59`; the manuscript’s reported utility values match the documented (not the actual code) default, blocking reproduction of headline QALYs/ICERs (TR-CONS-003).
4. Dependency lists are prose-only (no `DESCRIPTION/renv.lock`) and omit roughly 14 packages actually used in the repository, mostly DAG/statistics-only reports (TR-CONS-004).
5. `docs/README.md`’s render guidance (“render the whole report directory at once”) directly contradicts `AGENTS.md`’s explicit warning that economic reports must be rendered individually with `--to pdf` because their GFM pass fails (TR-CONS-005).

**Completeness gaps** (AdViSHE Parts A-D): no documented independent human review of the conceptual model (A1, TR-COMP-011), the input data (B1, TR-COMP-013), or the implemented code (C1, TR-COMP-015) was found anywhere in the corpus — the manuscript’s own Contributors section is a [To be completed] placeholder. No documentation of extreme-value/stress-testing (C2, TR-COMP-016) or of a predefined unit-test acceptance protocol (C4, TR-COMP-018) exists, though the underlying test scripts themselves do (`scripts/R/tests/`). No comparison of model outputs against other published decision models is reported (D2, TR-COMP-020); the manuscript self-describes as “the first” such evaluation, which is plausible but not stated explicitly as a scoping decision. Empirical validation (D4) is limited to a single narrative sentence comparing predicted 5-year OS to a published registry figure, with no quantified agreement statistic or dependent/independent classification (TR-COMP-022).

**Module 1 verdict rationale:** applying Module 1 verdict rule 3, the Model-A/B/C+TLR-scope contradiction and the `UTILITY_SOURCE` default contradiction are each (a) uncorrected within the documentation set and (b) block reproduction of decision-driving outputs, routing the module to **“Insufficiently transparent for review.”** `AGENTS.md` and the currently-rendered reports do consistently and adequately document the deployed single-model design; the defect is that the sibling `docs/README.md` and the manuscript were not updated to match it.

## 2.4 Face-validity council results

Not requested for this run. All CDA-AMC items 6-7, AdViSHE A1/B1/D1 substantive judgment, and TECH-VER Stage-5 clinical-expert-comparison rows are marked **Deferred (M2)** in the checklists below.

## 2.5 Black-box results

**Base case (BB-BASE, pass):** cost: control=20,867.2, crp=55,025.8, tmb\_braf=60,797.5 (EUR); QALY: control=1.41414, crp=1.38424, tmb\_braf=1.35303; LY: control=2.17082, crp=2.03532, tmb\_braf=2.03304.

Of 56 black-box tests, **43 passed** cleanly (base-case run; QALY cross-check against the model’s own internal aggregation; LY QALY; undiscounted discounted; QALY/LY ratio within the utility range; cohort-conservation, non-negativity, and absorbing-monotonicity for every trace and strategy; individual PFS/OS curve bounds and monotonicity for every strategy; utilities-forced-to-1/0; costs-forced-to-0; mortality-forced-to-~1; equalized-effectiveness; discount-rate extremes; shorter horizon; halving of effect difference; determinism re-run; all 4 PSA consistency checks (43-46); and all 9 unit-style tests of `validate_model_params/calculate_outcomes/extract_all_survival_probabilities`). The 10 fails and 3 not-tested items were:

| ID                  | Test                                        | Verdict | Severity                                 | Key result                                                                                                                                                  |
|---------------------|---------------------------------------------|---------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BB-PSM-ORD-control  | OS $\geq$ PFS, no crossing                  | Fail    | critical<br>(field)/minor<br>(magnitude) | PFS exceeds OS at 3/521 points, t=1-3wk only, max excess 0.0005 — near-t0 numerical noise                                                                   |
| BB-PSM-ORD-crp      | OS $\geq$ PFS, no crossing                  | Fail    | critical                                 | PFS exceeds OS at 251/521 points (48%); 49% of the entire horizon (t=267-520wk, max excess 0.0229) in the crp+ subgroup; progressed state forced to 0 there |
| BB-PSM-ORD-tmb_braf | OS $\geq$ PFS, no crossing                  | Fail    | critical                                 | PFS exceeds OS at 186/521 points (36%); 51% of the entire horizon (t=254-520wk, max excess 0.0145) in the tmb_braf+ subgroup                                |
| BB-PSM-WCHECK-crp   | Weighted curve reproduces precomputed field | Fail    | major                                    | 0.02pp prevalence drift between full-data (n=74) and complete-case (n=68) cohorts                                                                           |

| ID                     | Test                                        | Verdict | Severity | Key result                                                                                                                                                      |
|------------------------|---------------------------------------------|---------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BB-PSM-WCHECK-tmb_braf | Weighted curve reproduces precomputed field | Fail    | major    | 0.81pp prevalence drift (larger due to more missingness)                                                                                                        |
| BB-M0                  | Mortality=0: LY equals horizon              | Fail    | minor    | Undiscounted LY = 10.019y vs 10y expected (equal across arms) — 521-point grid sum with no half-cycle correction over-counts by ~1 cycle (~0.19%), ICER-neutral |
| BB-EQ2                 | Equal effect+cost: costs align              | Fail    | minor    | CRP arm +16.0 EUR residual — CRP test cost shares a field with routine blood-monitoring cost, cannot be independently zeroed                                    |
| BB-SW                  | Arm swap: outputs swap                      | Fail    | major    | QALYs/LYs swap exactly; costs do not ( $\pm 2,502$ EUR = NGS-vs-blood test-cost delta) — test cost keyed to biomarker name string, not data                     |
| BB-AC                  | Raise one arm's acquisition cost            | Fail    | minor    | Bumping the shared nivolumab cost field raises both crp and tmb_braf (intentional — same drug — but no per-strategy price hook exists)                          |

| ID              | Test                                           | Verdict    | Severity | Key result                                                                                                             |
|-----------------|------------------------------------------------|------------|----------|------------------------------------------------------------------------------------------------------------------------|
| BB-2X           | Double effect difference: ~2x incremental QALY | Fail       | minor    | Ratio 1.478, outside 30% band — compounded by probability-scale clamp saturation and the PSM-ORD crossing              |
| BB-D2Y          | $(1+r)^2$ spot check                           | Not-tested | info     | Per-cycle streams not synthesized in the harness; QALY/LY discounting already independently verified via BB-D0/D1      |
| BB-HCC          | Half-cycle correction toggle                   | Not-tested | info     | No dedicated HCC setter exists (grep confirms no HCC logic at all); functionally the same gap as BB-M0                 |
| BB-PSA-FALLBACK | PSA fallback rate                              | Not-tested | info     | 0/100 iterations fell back to base-case curves at $n=100$ — healthy, but not exercised at the documented full $n=5000$ |

## 2.6 White-box results

All 12 findings (performed by the orchestrator after the dispatched white-box subagent was cut off by a session limit — see §2.2):

| ID     | Test                                   | Verdict | Severity | Location                                                    |
|--------|----------------------------------------|---------|----------|-------------------------------------------------------------|
| WB-001 | Root cause of OS/PFS crossing          | Fail    | critical | 06_parametric_survival_anal.<br>model_fun.R:306-319,375-387 |
| WB-002 | Documented vs implemented distribution | Partial | major    | 06_parametric_survival_anal.                                |

| ID     | Test                                             | Verdict | Severity                     | Location                                               |
|--------|--------------------------------------------------|---------|------------------------------|--------------------------------------------------------|
| WB-003 | Crossing masked by<br>warning()+clamp            | Fail    | major                        | model_fun.R:306-312,375-382                            |
| WB-004 | Test cost keyed to name string                   | Fail    | major                        | calculate_outcomes.R:47-53                             |
| WB-005 | Prevalence drift root cause                      | Partial | minor                        | 03_biomarker_strategies.R:207_basecase_input_parameter |
| WB-006 | Discount formula and application                 | Pass    | info                         | model_fun.R:282-284                                    |
| WB-007 | Partition sums to 1 by construction              | Pass    | info                         | model_fun.R:318-320,385-387                            |
| WB-008 | No half-cycle correction;<br>endpoint over-count | Partial | minor                        | calculate_outcomes.R:28-32;<br>model_fun.R:282         |
| WB-009 | Forward/backward trace of u_p,<br>c_drug_nivo    | Pass    | info                         | calculate_outcomes.R:27-92                             |
| WB-010 | Treatment/monitoring costs accrue only to PF     | Pass    | minor<br>(documentation gap) | calculate_outcomes.R:35-63                             |
| WB-011 | PSA fallback path detectability                  | Partial | major                        | model_fun.R:236-259                                    |
| WB-012 | flexsurvreg returns survival, not hazard         | Pass    | info                         | prediction_functions.R                                 |

## 2.7 Coding-standards results

**CHAOS rubric scores:** Folder structure = **Adequate** (DARTH-style layout, human-readable names; not Advanced — no installable-package structure). Consistent style = **Adequate** (DARTH prefixes used correctly and consistently on domain objects; coverage gap on generic locals like `strategies/cost_matrix`). Coding documentation = **Adequate overall, 30.0% roxygen2 coverage repo-wide (62/207 functions)** — highly uneven: `functions/` 72.2% (effectively Advanced), `analysis/` 38.1%, `QMD/report` 17.5%, `QMD/vignettes` 9.4%, `archive//tests//technical_docs` 0%. Modularity = **Advanced** (fully modularized, numbered, prepare/run/analyze-separated pipeline with a clear call graph). Reproducibility = **Absent-to-Basic** (no `renv.lock`, no `DESCRIPTION`, no in-repo `README.md`; all 37 packages unpinned; confidential trial data intentionally excluded per data-governance policy, which is appropriately documented but still blocks outside reproduction).

All 14 findings:

| ID     | Test             | Verdict | Severity |
|--------|------------------|---------|----------|
| CS-001 | Folder structure | Pass    | info     |

| ID     | Test                                                      | Verdict    | Severity |
|--------|-----------------------------------------------------------|------------|----------|
| CS-002 | Consistent style /<br>DARTH prefix<br>coverage            | Partial    | minor    |
| CS-003 | Documentation /<br>roxygen2 coverage                      | Partial    | minor    |
| CS-004 | Modularity /<br>functions rubric                          | Pass       | info     |
| CS-005 | Reproducibility —<br>package &<br>dependencies / lockfile | Fail       | major    |
| CS-006 | Execution<br>reproducibility (static<br>assessment)       | Not-tested | info     |
| CS-007 | DRY violation —<br>duplicate<br>statistical-test helpers  | Partial    | minor    |
| CS-008 | Error-masking<br>patterns<br>(tryCatch/fallback)          | Partial    | minor    |
| CS-009 | <code>for (i in 1:n)</code><br>robustness idiom           | Partial    | minor    |
| CS-010 | Growing-object-in-<br>loop inefficiency                   | Partial    | info     |
| CS-011 | Naming drift<br>( <code>multi_model_cea.R</code> )        | Partial    | info     |
| CS-012 | Hidden <code>.GlobalEnv</code><br>dependency              | Partial    | minor    |
| CS-013 | Automated test<br>coverage (no <code>testthat</code> )    | Partial    | info     |
| CS-014 | Inventory<br>false-positive ( <code>pkg</code> )          | NA         | info     |

## 2.8 Cross-validation results

Not requested for this run. No comparator model was supplied or discoverable as reviewable comparator evidence. AdViSHE A2/D2 and the TECH-VER Stage-5 model-comparison row are marked **Deferred (M4)**.

## 2.9 External-validation results

Not requested for this run. No independent observed-event dataset was supplied. The manuscript's single narrative comparison of predicted 5-year OS against a published registry figure (Siegel et

al. 2025) is recorded under Module 1 (TR-COMP-022) but not independently validated here. Ad-ViSHE D3/D4/D4.A/D4.B substantive layers and the TECH-VER Stage-2 life-table / Stage-5 external-event-data rows are marked **Deferred (M5)**.

## 2.10 Completed checklists

### CDA-AMC Model Validation Tool

| Item | Description                                        | Verdict       | Evidence                                                                                         |
|------|----------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------|
| 1    | Target population aligns with decision problem     | Fail          | TR-CONS-001 (fail); TR-COMP-001 (pass, context)                                                  |
| 2    | Relevant population subgroups assessable           | Fail          | TR-CONS-001 (fail); TR-COMP-002 (partial)                                                        |
| 3    | Current treatment comparators represented          | Fail          | TR-CONS-001, TR-CONS-002 (fail); TR-COMP-003 (partial)                                           |
| 4    | Cost categories match perspective                  | Fail          | TR-CONS-003 (fail); TR-COMP-004 (pass, context)                                                  |
| 5    | Clinician/patient-important outcomes represented   | Partial       | TR-COMP-005 (no AE category documented)                                                          |
| 6    | Clinical experts reviewed structure                | Deferred (M2) | —                                                                                                |
| 7    | Structure consistent with prior models / justified | Deferred (M2) | (M1 doc-layer evidence: TR-CONS-002, TR-CONS-007 — see §2.3)                                     |
| 8    | State occupancy extractable                        | Pass          | BB-TR1-control, BB-QALY-XCHECK, BB-BASE                                                          |
| 9    | State-occupancy agrees with submitted evidence     | Not-tested    | No independently submitted state-occupancy dataset in repo distinct from the fitting data itself |

| Item | Description                                                 | Verdict    | Evidence                                                                                                                   |
|------|-------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------|
| 10   | Clinical-event counts extractable                           | Partial    | BB-TR2-control/crp/tmb_braf (death-state extraction confirmed); discrete progression-event counts not separately extracted |
| 11   | Clinical-event outputs agree with evidence                  | Not-tested | Same reason as item 9                                                                                                      |
| 12   | AE impacts isolable                                         | Fail       | TR-COMP-005 (no AE cost/outcome category exists)                                                                           |
| 13   | AE outputs agree with evidence                              | NA         | No AE outcomes modeled (see item 12)                                                                                       |
| 14   | Life-years available                                        | Pass       | BB-BASE, BB-LQ                                                                                                             |
| 15   | Mortality impact transparent                                | Pass       | BB-PSM-OS-control/crp/tmb_braf; WB-009                                                                                     |
| 16   | Mortality difference decomposed by cause                    | Not-tested | No decomposition report found                                                                                              |
| 16a  | Time in health states contributes                           | Not-tested | No decomposition report found                                                                                              |
| 16b  | Fatal clinical events contribute                            | Not-tested | No decomposition report found                                                                                              |
| 16c  | Fatal AE frequencies contribute                             | NA         | No AEs modeled (see item 12)                                                                                               |
| 16d  | Direct survival/death-risk effect contributes               | Not-tested | No decomposition report found                                                                                              |
| 17   | Main mortality driver identifiable                          | Not-tested | No decomposition report found                                                                                              |
| 18   | Mortality outputs agree with evidence                       | Not-tested | No submitted mortality-validation dataset in repo                                                                          |
| 19   | Model clarifies whether utilities differ by technology      | Pass       | WB-009, WB-010 (utilities are state-based, not technology-based, and documented)                                           |
| 20   | Largest driver among state/events/AE/mortality identifiable | Not-tested | biomarker_decomposition.qmd decomposes by biomarker, not by this axis                                                      |

| Item | Description                                               | Verdict    | Evidence                                                                                          |
|------|-----------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------|
| 21   | Extrapolated inputs/outputs flagged                       | Fail       | BB-PSM-ORD-crp/tmb_braf, WB-001<br>(extrapolation-region crossing not flagged beyond a warning()) |
| 22   | Horizon restrictable to observed-data period              | Fail       | Same evidence; no restriction capability found                                                    |
| 23   | Direct/indirect effects avoid double counting             | Not-tested | No dedicated test performed                                                                       |
| 24   | Surrogate-to-final-outcome relationship supported         | Not-tested | No submitted surrogate-validation evidence in repo                                                |
| 25   | Treatment-effect waning explorable / permanence justified | Not-tested | No waning mechanism or documentation found                                                        |
| 26   | Deterministic/single-run results accessible               | Fail       | TR-CONS-002/003/005/006 (fail); TR-COMP-024, BB-BASE (pass, context)                              |
| 27   | Inputs traceable to outcomes                              | Fail       | TR-CONS-002/003 (fail); TR-COMP-025/026, WB-005, WB-009 (partial/pass, context)                   |
| 28   | Excel macros limited to simulation/navigation             | NA         | Excel-exclusive                                                                                   |
| 29   | Equalized effectiveness -> equal QALYs                    | Pass       | BB-EQ1                                                                                            |
| 30   | Extreme effectiveness changes move QALYs                  | Not-tested | No dedicated extreme-effectiveness (non-doubling) test                                            |
| 31   | Small effectiveness changes move QALYs modestly           | Fail       | BB-2X (fail, minor); BB-HX (pass)                                                                 |
| 32   | Higher mortality risk lowers QALYs/LY                     | Not-tested | Only extremes (0/~1) tested, not moderate directional change                                      |
| 33   | Lower mortality risk raises QALYs/LY                      | Not-tested | Same reason as item 32                                                                            |
| 34   | Higher baseline event risk lowers QALYs                   | Not-tested | No dedicated test                                                                                 |

| Item | Description                                                | Verdict    | Evidence                                                                    |
|------|------------------------------------------------------------|------------|-----------------------------------------------------------------------------|
| 35   | Lower baseline event risk raises QALYs                     | Not-tested | No dedicated test                                                           |
| 36   | Zero mortality -> equal LY across arms                     | Partial    | BB-M0 (equal across arms confirmed; small systematic grid-count bias)       |
| 37   | Higher technology cost affects only relevant costs         | Fail       | WB-004, BB-AC (fail); WB-010 (pass, context)                                |
| 38   | Utility=1, no disutility -> QALY=LY                        | Pass       | BB-U1                                                                       |
| 39   | Zero discounting increases costs/QALYs                     | Pass       | BB-D0, WB-006                                                               |
| 40   | Higher discounting decreases costs/QALYs                   | Pass       | BB-D1, WB-006                                                               |
| 41   | Shorter horizon reduces totals                             | Partial    | BB-H1 (pass); WB-008 (grid-count/half-cycle gap)                            |
| 42   | Swapping inputs swaps results                              | Fail       | BB-SW, WB-004                                                               |
| 43   | Cross-technology cost/QALY correlations calculable         | Pass       | BB-PSA-43 (fallback-rate check at full n not-tested — see BB-PSA-FALLBACK)  |
| 44   | Cost correlations show expected pattern                    | Partial    | BB-PSA-44 (pass); WB-011 (fallback-checking completeness not verified)      |
| 45   | QALY correlations show expected pattern                    | Pass       | BB-PSA-45                                                                   |
| 46   | Deterministic and probabilistic results broadly consistent | Pass       | BB-PSA-46                                                                   |
| 47   | Backward trace (outputs to inputs)                         | Fail       | WB-001 (critical); WB-002 (partial), WB-007, WB-009, WB-012 (pass, context) |
| 48   | Forward trace (input through single run)                   | Fail       | WB-001 (critical), WB-003 (major)                                           |

| Item | Description                                                                    | Verdict | Evidence                                                                                          |
|------|--------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------|
| 49   | Error-masking /<br>hard-to-audit patterns<br>absent                            | Partial | CS-008<br>(well-commented but<br>non-hard-failing<br>fallback patterns);<br>WB-003, WB-011        |
| 50   | Hidden workbook<br>content absent<br>(R-equiv: no hidden<br>auto-loaded state) | Partial | CS-012 ( <code>.GlobalEnv</code><br>reliance in<br><code>resolve_best_distributions()</code> )    |
| 51   | VBA macros free of<br>hidden logic                                             | NA      | No R analogue                                                                                     |
| 52   | Macros/run<br>procedures don't reset<br>parameters (R-equiv:<br>determinism)   | Partial | BB-RE (pass);<br>WB-011 ( <code>&lt;&lt;-</code><br>fallback-flag pattern,<br>narrow but present) |
| 53   | Inputs centralized and<br>transparent                                          | Partial | TR-COMP-026                                                                                       |

**Rollup:** Pass 11 | Partial 9 | Fail 14 | NA 4 | Not-tested 17 | Deferred 2

## TECH-VER

| Stage / item                                                                                 | Verdict                | Evidence                                          |
|----------------------------------------------------------------------------------------------|------------------------|---------------------------------------------------|
| Stage 1 — input validation                                                                   | Pass                   | UNIT-VMP-01..05 (pass x5)                         |
| Stage 1 — prediction<br>extraction                                                           | Pass                   | UNIT-PRED-01                                      |
| Stage 1 — other pre-analysis<br>sub-tests (acquisition-<br>price/dosing/HR-import<br>checks) | Not-tested             | No dedicated evidence in this<br>run              |
| Stage 2 — cohort sums to<br>cohort size                                                      | Pass                   | BB-TR1-<br>control/crp/tmb_braf                   |
| Stage 2 — occupancy in [0,1]                                                                 | Pass                   | BB-TR0-<br>control/crp/tmb_braf                   |
| Stage 2 — absorbing-state<br>monotonicity                                                    | Pass                   | BB-TR2-<br>control/crp/tmb_braf                   |
| Stage 2 — survival curve<br>bounds/monotonicity                                              | Pass                   | BB-PSM-PFS/OS-<br>control/crp/tmb_braf (x6)       |
| Stage 2 — <b>PSM partition:<br/>OS &gt;= PFS</b>                                             | <b>Fail (critical)</b> | BB-PSM-ORD-crp,<br>BB-PSM-ORD-tmb_braf,<br>WB-001 |
| Stage 2 — weighted-curve<br>consistency                                                      | Fail                   | BB-PSM-WCHECK-crp,<br>BB-PSM-WCHECK-tmb_braf      |

| Stage / item                                                                                                                                           | Verdict         | Evidence                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------|
| Stage 2 — utilities=1/0, costs=0                                                                                                                       | Pass            | BB-U1, BB-U0, BB-C0                                                                             |
| Stage 2 — mortality=0                                                                                                                                  | Fail (minor)    | BB-M0                                                                                           |
| Stage 2 — mortality extreme                                                                                                                            | Pass            | BB-M1                                                                                           |
| Stage 2 — equal-arms                                                                                                                                   | Pass            | BB-EQ1                                                                                          |
| Stage 2 — equal-arms + costs                                                                                                                           | Fail (minor)    | BB-EQ2                                                                                          |
| Stage 2 — arm-swap                                                                                                                                     | Fail            | BB-SW, WB-004                                                                                   |
| Stage 2 — acquisition cost isolation                                                                                                                   | Fail (minor)    | BB-AC                                                                                           |
| Stage 2 — cost/QALY per-cycle calculation                                                                                                              | Pass            | UNIT-CO-01..03, BB-QALY-XCHECK                                                                  |
| Stage 2 — other sub-tests (DES sampling, decision-tree nodes, CRN, correlated characteristics, tunnel states, inflation, population-model incidence=0) | NA / Not-tested | Structurally inapplicable to a non-DES, non-decision-tree, non-population PSM, or not exercised |
| Stage 3 — LY>=QALY, undisc>=disc, QALY/LY in range                                                                                                     | Pass            | BB-LQ, BB-UD, BB-QLU                                                                            |
| Stage 3 — discount=0 / discount up                                                                                                                     | Pass            | BB-D0, BB-D1, WB-006                                                                            |
| Stage 3 — shorter horizon                                                                                                                              | Pass            | BB-H1                                                                                           |
| Stage 3 — halving effect difference                                                                                                                    | Pass            | BB-HX                                                                                           |
| Stage 3 — cost-weighting to health states                                                                                                              | Pass            | WB-010                                                                                          |
| Stage 3 — half-cycle correction / endpoint over-count                                                                                                  | Partial         | WB-008, BB-M0 (no HCC toggle exists; ~0.19% systematic, ICER-neutral bias)                      |
| Stage 3 — doubling effect difference                                                                                                                   | Fail (minor)    | BB-2X                                                                                           |
| Stage 3 — $(1+r)^2$ spot check                                                                                                                         | Not-tested      | BB-D2Y                                                                                          |
| Stage 4 — PSA cost/QALY bounds and cross-strategy correlation                                                                                          | Pass            | BB-PSA-43, BB-PSA-44, BB-PSA-45                                                                 |
| Stage 4 — deterministic vs PSA-mean consistency                                                                                                        | Pass            | BB-PSA-46                                                                                       |
| Stage 4 — PSA fallback rate                                                                                                                            | Not-tested      | BB-PSA-FALLBACK (0% at n=100; not exercised at full n=5000)                                     |
| Stage 4 — PSA fallback-checking completeness                                                                                                           | Partial         | WB-011                                                                                          |

| Stage / item                                                                                      | Verdict       | Evidence                          |
|---------------------------------------------------------------------------------------------------|---------------|-----------------------------------|
| Stage 4 — OWSA<br>bounds/distribution-choice<br>appropriateness, CEAC/VOI<br>internal consistency | Not-tested    | No dedicated evidence in this run |
| Stage 5 — determinism (run twice, same answer)                                                    | Pass          | BB-RE                             |
| Stage 5 — DRY /<br>duplicate-code hygiene                                                         | Partial       | CS-007                            |
| Stage 5 — error-masking /<br>silent-fallback patterns                                             | Partial       | CS-008, WB-011                    |
| Stage 5 — loop-idiom<br>robustness                                                                | Partial       | CS-009                            |
| Stage 5 —<br>growing-object-in-loop<br>inefficiency                                               | Partial       | CS-010                            |
| Stage 5 — naming drift /<br>dead-code hygiene                                                     | Partial       | CS-011                            |
| Stage 5 — hidden <code>.GlobalEnv</code><br>dependency                                            | Partial       | CS-012                            |
| Stage 5 — diagnostic-<br>cost/strategy-identity coupling                                          | Fail          | WB-004, BB-SW                     |
| Stage 5 — comparison vs other<br>models/published analyses                                        | Deferred (M4) | —                                 |
| Stage 5 — comparison vs<br>clinical expert knowledge                                              | Deferred (M2) | —                                 |
| Stage 5 — comparison vs<br>external event data                                                    | Deferred (M5) | —                                 |

**Rollup:** Pass 27 | Partial 8 | Fail 7 | NA 0 | Not-tested 5 | Deferred 3

## AdViSHE

| Item | Description                                                 | Verdict | Evidence                                                                           |
|------|-------------------------------------------------------------|---------|------------------------------------------------------------------------------------|
| A1   | Conceptual model<br>reviewed by qualified<br>experts        | Fail    | TR-COMP-011 (no<br>named panel;<br>Contributors section is<br>a placeholder)       |
| A2   | Conceptual model<br>compared with other<br>models/textbooks | Partial | TR-COMP-012 (M1<br>reporting layer only;<br>M4 substantive<br>comparison Deferred) |
| B1   | Input data reviewed<br>by qualified experts                 | Fail    | TR-COMP-013                                                                        |

| Item | Description                                         | Verdict | Evidence                                                                                                                                                                          |
|------|-----------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B2   | Statistical fit tests for regression-derived inputs | Partial | TR-COMP-014 (pass, M1 reporting); WB-002 (partial — documented “gamma for both” claim does not match implemented AIC-selected distributions); WB-012 (pass)                       |
| C1   | External review by independent modelling experts    | Partial | Meta rule: the only external computerized-model review found is this ValidateHE AI-assisted run itself; TR-COMP-015 confirms no prior documented independent human review         |
| C2   | Extreme-value testing                               | Fail    | TR-COMP-016 (no documented extreme-value test protocol, though M3 executed extensive ad hoc extreme-value tests: BB-U1/U0/C0/M0/M1/EQ1/EQ2/D0, mostly passing with 2 minor fails) |
| C3   | Testing of traces                                   | Fail    | WB-001 (critical), BB-PSM-ORD-crp/tmb_braf (critical); TR-COMP-017 (M1 reporting, partial)                                                                                        |

| Item | Description                                    | Verdict       | Evidence                                                                                                                                                                                    |
|------|------------------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C4   | Unit testing under predefined protocol         | Partial       | TR-COMP-018 (partial — no documented acceptance protocol); UNIT-VMP-01..05, UNIT-CO-01..03, UNIT-PRED-01 (all pass — the scratch-generated test harness itself is the protocol per routing) |
| D1   | Outputs judged reasonable by qualified experts | Partial       | TR-COMP-019 (M1 reporting; M2 substantive judgment Deferred)                                                                                                                                |
| D2   | Outputs compared with other models             | Fail          | TR-COMP-020                                                                                                                                                                                 |
| D3   | Outputs compared under alternative input data  | Pass          | TR-COMP-021 (M1 reporting layer — OWSA/scenario/UTILITY_SOURCE toggle all documented; M5 substantive re-run Deferred)                                                                       |
| D4   | Outputs compared with empirical data           | Partial       | TR-COMP-022                                                                                                                                                                                 |
| D4.A | Dependent validation                           | Deferred (M5) | Substantive comparison against fitting-data sources owned by M5                                                                                                                             |
| D4.B | Independent validation                         | Partial       | TR-COMP-022 (single narrative registry comparison, no quantified agreement statistic)                                                                                                       |
| E1   | Other validation techniques                    | Pass          | BB-RE (determinism re-run recorded as the additional technique performed); TR-COMP-023 (M1: no additional technique claimed in documentation)                                               |

**Rollup:** Pass 2 | Partial 7 | Fail 5 | NA 0 | Not-tested 0 | Deferred 1

## CHAOS v0.9

| Rubric                         | Verdict | Evidence                                                                                                                                                                  |
|--------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Starting point / README        | Partial | TR-COMP-006 (four required elements nominally present, but the README's own model description materially disagrees with <code>AGENTS.md</code> /deployed code)            |
| Run guidance                   | Pass    | TR-COMP-007 (Advanced when <code>AGENTS.md</code> + README are combined)                                                                                                  |
| License                        | Pass    | TR-COMP-008 (MIT, correctly referenced)                                                                                                                                   |
| Package & dependencies         | Fail    | CS-005 (no <code>renv.lock</code> /DESCRIPTION anywhere; all 37 packages unpinned); TR-COMP-009 (partial, M1 doc layer)                                                   |
| Model version (history)        | Partial | TR-COMP-010 (per-document version footers + issue-numbered technical docs, but no unified changelog; live snapshot artifact currently empty)                              |
| Folder structure               | Pass    | CS-001 (DARTH-style layout, Adequate)                                                                                                                                     |
| Consistent style               | Partial | CS-002 (DARTH prefixes correct where used; coverage gap on generic locals); CS-011                                                                                        |
| Coding documentation           | Partial | CS-003 (30.0% roxygen2 coverage repo-wide, very uneven by folder); CS-013                                                                                                 |
| Functions/modules (Modularity) | Partial | CS-004 (pass, Advanced pipeline modularization); CS-012 (partial, one hidden- <code>.GlobalEnv</code> -dependency instance)                                               |
| Execution reproducibility      | Fail    | CS-005 (no lockfile); CS-006 (not-tested — static assessment only; confidential trial data intentionally excluded from public repo per documented data-governance policy) |

**Rollup:** Pass 3 | Partial 5 | Fail 2 | NA 0 | Not-tested 0 | Deferred 0

## Appendix A — Recommended fixes (NOT applied)

**ValidateHE modified nothing.** The repository under review is byte-for-byte unchanged (integrity hash verified before and after this review: 4402F766BEF05DFF926D4126218006DA97A7C9C0FEF411956BB6E6C746A01068 / 4402F766BEF05DFF926D4126218006DA97A7C9C0FEF411956BB6E6C746A01068). The changes below are recommendations for the model’s maintainers, in descending order of severity.

| ID     | File:line                                                                                                  | Category              | Severity | Issue                                                                                                 | Recommended change                                                                                                                                                                                                                      | Checklist item(s)                                |
|--------|------------------------------------------------------------------------------------------------------------|-----------------------|----------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| WB-001 | scripts/R/analysis/fit06-parametric-survival.R:104-156;<br>scripts/R/functions/model_fun.R:306-319,375-387 | WIS-06 (verification) | Critical | OS=O, PFS fit as independent AIC-selected models with no OS>=PFS ordering constraint on extrapolation | Fit PFS and the PFS-to-OS gap jointly so ordering holds by construction (or otherwise constrain extrapolated OS>=PFS); at minimum elevate the pmin() clamp to a reported diagnostic and quantify the QALY/cost impact before publishing | CDA-AMC 47, 48; TECH-VER Stage 2; AdViSHE B2, C3 |

| ID                  | File:line                                                                                                                                                     | Category                     | Severity | Issue                                                                                                   | Recommended change                                                                                                                                                        | Checklist item(s)                      |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| BB-PSM-ORD-crp      | scripts/R/functions/Black-box-model_fit.R:376-382;<br>scripts/R/functions/prediction_functions.R:190;<br>scripts/R/analysis/06_parametric_survival_analysis.R | Black-box (verification)     | Critical | PFS exceeds OS at 49% of the full 10y horizon for the crp+ subgroup; progressed state forced to 0 there | Same as WB-001; alternatively cap extrapolation at last observed follow-up and document a transparent tail assumption (CDA-AMC 21/22)                                     | TECH-VER 3.2 PSM partition; AdViSHE C3 |
| BB-PSM-ORD-tmb_braf | scripts/R/functions/Black-box-model_fit.R:411;<br>scripts/R/functions/prediction_functions.R:190                                                              | Black-box (verification)     | Critical | PFS exceeds OS at 51% of the full 10y horizon for the tmb_braf+ subgroup (a headline economic strategy) | Same as WB-001/BB-PSM-ORD-crp                                                                                                                                             | TECH-VER 3.2 PSM partition; AdViSHE C3 |
| TR-COMP-002         | S02 docs/README (documentation)<br>'Biomarker Strategies Evaluated';<br>R01 Methods panel                                                                     | Transparency (documentation) | Major    | README/manuscript describe a 3rd TLR economic subgroup no longer in code                                | Manuscript README/manuscript TLR-strategy/Model-A-C material as superseded or historical; state which subgroup set the deployed model evaluates in one canonical location | CDA-AMC 2                              |

| ID          | File:line                                                 | Category                               | Severity | Issue                                                                                       | Recommended change                                                                                                          | Checklist item(s) |
|-------------|-----------------------------------------------------------|----------------------------------------|----------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------|
| TR-COMP-003 | S02<br>docs/README<br>'Key Features';<br>R01 Table<br>4-6 | Transparency (document-<br>ation)      | Major    | README advertises a 4th comparator arm and Model A/B/C feature not present in code          | Reconcile which comparator set is current, or date/label the manuscript's broader framework as prior/exploratory            | CDA-AMC<br>3      |
| TR-COMP-005 | docs_extracted/Transparency_Maps.txt<br>Table 4           | Transparency Maps (document-<br>ation) | Major    | No AE/toxicity outcome or cost category documented anywhere, no stated rationale            | Add a brief scope statement on whether AE costs/disutilities were considered and why excluded if so                         | CDA-AMC<br>5      |
| TR-COMP-011 | R01 'Contributors' section<br>(placeholder)               | Transparency (document-<br>ation)      | Major    | No named expert panel/review of conceptual model; Contributors section is [To be completed] | Complete the Contributors section or add a 'Conceptual model validation' subsection stating this was not formally performed | AdViSHE<br>A1     |

| ID          | File:line                                             | Category                     | Severity | Issue                                                                                                      | Recommended change                                                                                                  | Checklist item(s)                    |
|-------------|-------------------------------------------------------|------------------------------|----------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| TR-COMP-013 | R01<br>'Clinical data source' section                 | Transparency (documentation) | Major    | No independent input-data review process documented                                                        | Document whether any co-author independently reviewed input-data extraction/coding, or state this was not performed | AdViSHE B1                           |
| TR-COMP-016 | not found in reviewed materials                       | Transparency (documentation) | Major    | No documentation of extreme-value/boundary condition testing                                               | Add a documented extreme-value test suite and report results/location                                               | AdViSHE C2; CDA-AMC 29-41 (adjacent) |
| TR-COMP-022 | R01<br>'Parametric survival models' results paragraph | Transparency (documentation) | Major    | Single narrative empirical comparison with no quantified statistic or dependent/independent classification | Add a short external/empirical validation subsection quantifying the comparison and classifying it                  | AdViSHE D4, D4.B                     |

| ID          | File:line                                                                                                   | Category                     | Severity | Issue                                                                                           | Recommended change                                                                                                                   | Checklist item(s)     |
|-------------|-------------------------------------------------------------------------------------------------------------|------------------------------|----------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| TR-COMP-026 | S01<br>AGENTS.md<br>'Critical Global Variables'<br>table;<br>docs_extracted/input_parameters.txt<br>Table 3 | Transparency (documentation) | Major    | AGENTS.md documented UTIL-ITY_SOURCE default disagrees with what actually renders               | Correct AGENTS.md's default value (or Evert code), and state explicitly which UTIL-ITY_SOURCE produced the reported results          | CDA-AMC s3-equivalent |
| TR-CONS-001 | docs/README.md 17,192;<br>scripts/R/functions/model_configs.R:28,31,117,118                                 | Transparency (documentation) | Major    | README documents TLR as one of three economic biomarker strategies; code explicitly excludes it | Edit README's 'Biomarker Strategies Evaluated' section to list only CRP and TMB/BRAF; note TLR is clinical-only, mirroring AGENTS.md | CDA-AMC 1, 2, 3       |

| ID          | File:line                                                                                                                                          | Category                     | Severity | Issue                                                                                                                                                 | Recommended change                                                                                                                                                       | Checklist item(s)    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| TR-CONS-002 | docs/README.transp-195;<br>scripts/R/functions/multi_model_cea.R;<br>scripts/R/functions/model_configs.R:28,44-63;<br>manuscript:110-150           | Transparency (documentation) | Major    | README/manuscript describe Model A/B/C + TLR strategy; no such code exists (structural absence, not just untested claim)                              | Rescript Model A/B/C/TLR claims from README, rename/gut multi_model_cea.R to match its actual single-model scope, note manuscript's A/C/TLR specifications as historical | CDA-AMC 3, 7, 26, 27 |
| TR-CONS-003 | AGENTS.md (Critical Global Variables table);<br>02_setup_and_global_variables.R:59;<br>07_basecase_input_parameters.R:63-64;<br>manuscript:102,174 | Transparency (documentation) | Major    | Documented UTIL-SOURCE default (0, IPD) disagrees with code's actual default (1, CORRECT); manuscript matches the documented, not the actual, default | Update AGENTS.md's default (or revert code); state explicitly which value generated the reported results, with corresponding cache filenames                             | CDA-AMC 4, 26, 27    |

| ID                     | File:line                                                                   | Category                         | Severity | Issue                                                                                                                                        | Recommended change                                                                                                                                                                     | Checklist item(s)                      |
|------------------------|-----------------------------------------------------------------------------|----------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| BB-PSM-WCHECK-crp      | scripts/R/analysis/03_biomarkers/01_verification/prediction_functions.R:164 | Black-box biomarker verification | Major    | Major strategies 0.0224; prevalence drift between full-data and complete-case cohorts affects the biomarker decomposition.qmd ‘Weighted’ row | Thread a single prevalence value into generate_populationC3averaged_predictions()                                                                                                      | TECH-VER 3.2 PSM partition; AdViSHE    |
| BB-PSM-WCHECK-tmb_braf | scripts/R/analysis/03_biomarkers/01_verification/prediction_functions.R:164 | Black-box biomarker verification | Major    | Major strategies 0.0126; prevalence drift (40x larger than CRP’s)                                                                            | Same fix as BB-PSM-WCHECK-crp; more urgent instance of the same defect                                                                                                                 | TECH-VER 3.2 PSM partition; AdViSHE C3 |
| BB-SW                  | scripts/R/functions/Black-box/01_verification/53                            | Black-box biomarker verification | Major    | Major outcomes.R:164 Diagnostic-test cost attached to strategy NAME not DATA; swapping curves doesn’t swap costs                             | Refactor calculate_outcomes.R to look up biomarker test cost via a params-level mapping rather than an if/else on the name, or document the coupling explicitly next to ALL_BIOMARKERS | TECH-VER 3.2 Rm-swap; CDA-AMC 42       |

| ID     | File:line                                                       | Category                                    | Severity | Issue                                                                                   | Recommended change                                                                                                                                  | Checklist item(s)                        |
|--------|-----------------------------------------------------------------|---------------------------------------------|----------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| WB-002 | scripts/R/analysis/06-parametric_survival/analysis.R:101<br>156 | Wlist/06-parametric_survival (verification) | Major    | states gamma selected for both endpoints; code auto-selects by AIC over 9 distributions | Either hard-code the documented gamma family or update documentation to state distributions are auto-selected and report which were actually chosen | CDA-AMC 47; TECH-VER Stage 2; AdViSHE B2 |
| WB-003 | scripts/R/function/whiteboard_model_fit.R:306-312,375-382       | Whiteboard_model_fit (verification)         | Major    | OS/PFS crossing surfaced only as warning(), easily lost in batch/PSA runs               | Fix the ordering at the source (WB-001); while the clamp remains, record clamped fraction and QALY/cost delta per strategy/iteration                | CDA-AMC 48; TECH-VER Stage 5; AdViSHE C3 |
| WB-004 | scripts/R/function/whiteboard_calculate_costs.R:53              | Whiteboard_calculate_costs (verification)   | Major    | NGS/CRP test costs keyed to biomarker name string, not data carried by the strategy     | Carry each strategy's diagnostic-test cost as a data field on the strategy/parameter object rather than branching on a hard-coded name              | CDA-AMC 37, 42; TECH-VER Stage 5         |

| ID          | File:line                                            | Category                       | Severity | Issue                                                                                             | Recommended change                                                                                                                                     | Checklist item(s)                             |
|-------------|------------------------------------------------------|--------------------------------|----------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| WB-011      | scripts/R/functions/white-box-model_verification_259 | White-box model (verification) | Major    | PSA fallback-to-base-case-curve path depends on every caller checking the fallback_used attribute | Confirm 12_PSA.R and other callers hard-check fallback_used/n_failed and exclude/report affected iterations; consider failing loudly above a threshold | CDA-AMC 44, 52 (R-equiv); TECH-VER Stage 5    |
| CS-005      | model/(repo root)                                    | Coding standards               | Major    | No renv.lock, DESCRIPTION, or in-repo README.md store/keep all 37 packages unpinned               | Add renv.lock pinning all package versions; re-README.md with version list and install guidance; add a minimal DESCRIPTION                             | CHAOS Package & dependencies, Reproducibility |
| TR-COMP-006 | S02 docs/README 'Contact'/'Key Features'             | Transparency (documentation)   | Minor    | Bare GitHub username as contact; model description disagrees with AGENTS.md                       | Add a real contact email/ORCID; update README's model description to match the deployed model                                                          | CHAOS Starting point/README                   |

| ID          | File:line                                                               | Category                          | Severity | Issue                                                                                                   | Recommended change                                                                                                                                 | Checklist item(s)                       |
|-------------|-------------------------------------------------------------------------|-----------------------------------|----------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| TR-COMP-009 | S02<br>docs/README-<br>'Prerequisite-<br>sites';<br>environ-<br>ment.md | Transparency (documen-<br>tation) | Minor    | Packages listed by name only, no versions, no DESCRIPTION/renv.lockable)                                | Add an<br>renv.lock<br>(or<br>minimally a<br>versioned<br>package<br>table)                                                                        | CHAOS<br>Package &<br>dependen-<br>cies |
| TR-COMP-010 | docs_extracted/<br>model_b_comp/                                        | Transparency (documen-<br>tation) | Minor    | No unified changelog/semver scheme; live snapshot artifact currently empty                              | Add a sin-<br>gle top-level<br>CHANGELOG<br>re-run<br>15_save_snapshot.R                                                                           | CHAOS<br>Model<br>version<br>(history)  |
| TR-COMP-012 | R01<br>'Research<br>in context'<br>section; refs<br>27, 28              | Transparency (documen-<br>tation) | Minor    | Other<br>mCRC<br>models<br>cited as<br>parameter<br>sources<br>only, not as<br>structural<br>comparison | Add a<br>paragraph<br>explicitly<br>comparing<br>the 3-state<br>PSM<br>structure<br>against<br>comparable<br>published<br>models                   | AdViSHE<br>A2                           |
| TR-COMP-017 | scripts/R/analy-<br>figs/traces_*.pdf                                   | Transparency (documen-<br>tation) | Minor    | Trace plots generated but not explicitly framed as a validation activity with a reported outcome        | Add a<br>paragraph<br>stating<br>trace plots<br>were<br>inspected<br>for the doc-<br>umented<br>constraints<br>and summa-<br>rizing the<br>outcome | AdViSHE<br>C3                           |

| ID          | File:line                                                       | Category                           | Severity | Issue                                                                                  | Recommended change                                                                                                 | Checklist item(s) |
|-------------|-----------------------------------------------------------------|------------------------------------|----------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------|
| TR-COMP-018 | S01<br>AGENTS.md<br>'Test Files';<br>scripts/R/tests/*.R        | Transparency (document-<br>tation) | Minor    | Test scripts named/described but no predefined acceptance protocol or pass/fail report | Add a 'Test protocol and results' subsection with pass/fail criteria and current status per script                 | AdViSHE C4        |
| TR-COMP-019 | R01<br>'Parametric survival models' results paragraph           | Transparency (document-<br>tation) | Minor    | Clinical plausibility statement not framed as a distinct expert-judgment exercise      | Add an explicit statement identifying which co-author(s) reviewed outputs for plausibility in a formal capacity    | AdViSHE D1        |
| TR-COMP-020 | not found in reviewed materials                                 | Transparency (document-<br>tation) | Minor    | No explicit statement that no comparator model exists                                  | Add a sentence stating this explicitly rather than leaving the comparison silently absent                          | AdViSHE D2        |
| TR-COMP-025 | S01<br>AGENTS.md<br>'Key Functions';<br>model_fun.R<br>comments | Transparency (document-<br>tation) | Minor    | No single consolidated input-to-output traceability table/diagram                      | Add a table/diagram linking each l_params_base field to the model_fun()/calculate_outcomes() steps that consume it | CDA-AMC 27        |

| ID          | File:line                                                                   | Category                     | Severity | Issue                                                                                        | Recommended change                                                                                               | Checklist item(s) |
|-------------|-----------------------------------------------------------------------------|------------------------------|----------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------|
| TR-CONS-004 | docs/README-Installation; AGENTS.md<br>Package Installation; environment.md | Transparency (documentation) | Minor    | Prose package lists omit ~14 packages actually used (mostly DAG/statistical reports)         | Generate a DESCRIPTION or renv.lock from the ambient library covering all detected packages                      | CDA-AMC 26        |
| TR-CONS-005 | docs/README-173; AGENTS.md<br>rendering-guidance section                    | Transparency (documentation) | Minor    | README's 'render whole directory' instruction contradicts AGENTS.md's per-report warning     | Remove the 'render all reports at once' instruction from README, or replace with AGENTS.md's per-report guidance | CDA-AMC 26        |
| TR-CONS-006 | scripts/QMD/Technical_docs/Model_b_conceptual.pdf                           | Transparency (documentation) | Minor    | Technical doc's 'Current Model B' section is itself stale relative to the June 2026 refactor | Add a superseded/deprecated banner pointing to model_configs.R as the current source of truth                    | CDA-AMC 26        |
| TR-CONS-007 | scripts/R/functions/participation46<br>docs/README-173                      | Transparency (documentation) | Minor    | File's own header says single-model-only; filename and README imply multi-model comparison   | Rename multi_model_7ea.R to single_model_cea.R (or similar) and update README's file-listing description         | CDA-AMC 26        |

| ID                 | File:line                                                                                                       | Category                 | Severity | Issue                                                                                                 | Recommended change                                                                                                | Checklist item(s)                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------|----------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| BB-PSM-ORD-control | scripts/R/functions_black-box_model_fit.R:306-312                                                               | Black-box (verification) | Minor    | PFS exceeds OS at 3/521 points, t=1-3wk only, max excess 0.0005                                       | Likely early-curve numerical noise; low priority relative to crp/tmb_braf; monitor if it grows after data updates | TECH-VER 3.2 PSM partition; AdViSHE C3                |
| BB-M0              | scripts/R/functions_black-box_model_fit.R:97-98; calculate_outcomes.R:28-32; 02_setup_and_global_variables.R:22 | Black-box (verification) | Minor    | 521-point grid sum over a 520-week horizon over-counts LY by ~0.19% (equal across arms, ICER-neutral) | Apply a half-cycle correction, drop the 521st point, or explicitly document the ~10.019y actual horizon           | TECH-VER 3.2 mortality=0, 3.3 LY==horizon; CDA-AMC 36 |
| BB-EQ2             | scripts/R/functions_black-box_calculate_outcomes.R:53                                                           | Black-box (verification) | Minor    | CRP test cost shares a field with routine blood-monitoring cost; can't be independently zeroed        | Add a dedicated c_test_CRP parameter distinct from the shared c_test_blood field                                  | TECH-VER 3.2 equal-arms+costs                         |
| BB-AC              | scripts/R/functions_black-box_calculate_outcomes.R:41                                                           | Black-box (verification) | Minor    | No way to vary nivolumab acquisition cost for one biomarker-guided strategy only                      | Add a per-strategy price multiplier if a single-pathway pricing scenario is ever needed                           | TECH-VER 3.2 acquisition cost; CDA-AMC 37             |

| ID     | File:line                                                                            | Category                                 | Severity | Issue                                                                  | Recommended change                                                                                                                        | Checklist item(s)                 |
|--------|--------------------------------------------------------------------------------------|------------------------------------------|----------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| BB-2X  | validatehe_test/validatehe_model_fun.R:306-312/376-382/404-411                       | Blatant box_hallucination (verification) | Minor    | Doubling effect-size test ratio (1.478) outside the 30% band around 2x | Not a standalone defect; scale the hazard/coefficient rather than the probability curve for a cleaner test; fixing WB-001 would also help | TECH-VER 3.3 doubling; CDA-AMC 31 |
| WB-005 | scripts/R/analysis/03_biomarker_strategy/07_basecase_verification_parameters.R:26-36 | Wishful thinking (verification)          | Minor    | Update prevalence definitions (full-data vs complete-case) can drift   | Compute prevalence once from a single cohort definition; derive both the pos/neg weighting and the weighted curves from it                | CDA-AMC 27; TECH-VER Stage 5      |
| WB-008 | scripts/R/function/whisper/whisper_calculate_model_fun.R:28-32-38-42 (verification)  | Wishful thinking (verification)          | Minor    | half-cycle correction; ~0.19% systematic LY/QALY over-count            | Apply a half-cycle correction or document the ~0.19% over-count as an accepted approximation                                              | CDA-AMC 41; TECH-VER Stage 3      |

| ID     | File:line                                                                           | Category                         | Severity | Issue                                                                                                                              | Recommended change                                                                                        | Checklist item(s)            |
|--------|-------------------------------------------------------------------------------------|----------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------|
| WB-010 | scripts/R/functions/whisper_calculate_min_costs.R:35-63                             | Writing standards (verification) | Minor    | Costs accrue only to progression-free occupancy; assumption not explicitly documented                                              | Document the assumption that treatment/monitoring costs cease at progression                              | CDA-AMC 37; TECH-VER Stage 3 |
| CS-002 | scripts/R/functions/clinical_gsa_function_parameters.R:207-3; model_configs.R:28,31 | Coding standards                 | Minor    | DARTH prefix coverage gap on generic locals (strategies, cost_matrix, ALL_STRATEGIES)                                              | Extend DARTH prefixing to local vectors/matrices/data frames incrementally, no changes                    | CHAOS Consistent style       |
| CS-003 | scripts/R/functions/clinical_gsa_analysis_tests.R:72.2%, 38.1%, 0%                  | Coding standards                 | Minor    | roxygen2 coverage only 30.0% repo-wide, very uneven                                                                                | Add roxygen2 (or a structured comment block) to analysis-script helpers and scripts/R/tests/*.R           | CHAOS Documentation          |
| CS-007 | scripts/QMD/Coding/clinical_effectiveness/forrest-plot.qmd:60                       | Coding standards                 | Minor    | Function names defined more than once across re-ports/vignettes (currently in sync, but a future fix could silently not propagate) | Move shared statistical-test helpers into a shared scripts/R/functions/*.R file sourced by both documents | TECH-VER Stage 5             |

| ID     | File:line                                                                    | Category                 | Severity | Issue                                                                                       | Recommended change                                                                                                        | Checklist item(s)                  |
|--------|------------------------------------------------------------------------------|--------------------------|----------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| CS-008 | scripts/R/functions/Clouds/model_fun.R:205,255-259;<br>08_sampling.R:117-134 | Clouds/g standards       | Minor    | Well-engineered but warning()-only fallback patterns depend on every caller checking a flag | Confirm every caller of model_fun() in PSA mode inspects fall-back_used/n_failed and excludes/reports affected iterations | TECH-VER Stage 5                   |
| CS-009 | scripts/R/functions/Clouds/gvppi_fun.R:74, and 7 more locations              | Clouds/g standards       | Minor    | for(i in 1:n) idiom breaks on zero-length input (latent, not currently triggered)           | Replace with for(i in seq_len(n)) throughout; mechanical, low-risk                                                        | TECH-VER Stage 5                   |
| CS-012 | scripts/R/analysis/08_sampling.R:189                                         | Clouds/g standards       | Minor    | resolve_best_models from .GlobalEnv rather than as a parameter                              | Pass distributions() models as an explicit parameter with a backward-compatible default                                   | CHAOS Modularity; TECH-VER Stage 5 |
| BB-D2Y | validatehe_test.R                                                            | Black-box (verification) | Info     | Per-cycle streams not synthesized in the harness for a direct (1+r)^2 spot check            | Low priority given BB-D0/BB-U1 already independently verify the discount arithmetic                                       | TECH-VER 3.3 (1+r)^2 spot check    |

| ID                  | File:line                                                            | Category                         | Severity | Issue                                                                                                                              | Recommended change                                                                                                                                             | Checklist item(s)                                       |
|---------------------|----------------------------------------------------------------------|----------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| BB-HCC              | model_fun.R;<br>calcu-<br>late_outcomeest10n                         | Black-box<br>(verifica-<br>tion) | Info     | No<br>half-cycle-<br>correction<br>toggle<br>exists at all<br>in this<br>weekly-<br>cycle PSM                                      | See<br>BB-M0's<br>recom-<br>mended fix;<br>not a defect<br>per se at<br>weekly-<br>cycle length<br>except for<br>the grid-<br>counting<br>artifact             | TECH-<br>VER 3.3<br>half-cycle<br>correction            |
| BB-PSA-<br>FALLBACK | model_fun.R<br>PSA path                                              | Black-box<br>(verifica-<br>tion) | Info     | 0% fallback<br>at n=100;<br>not<br>exercised at<br>the docu-<br>mented full<br>n=5000                                              | Re-check<br>periodically<br>at full n; a<br>creeping<br>fallback<br>rate would<br>silently<br>understate<br>PSA-driven<br>uncertainty                          | CDA-AMC<br>43; TECH-<br>VER 3.4<br>Monte<br>Carlo error |
| CS-001              | scripts/R/,<br>scripts/QMD/                                          | Coding<br>standards              | Info     | Not an<br>installable<br>R package<br>(no<br>DESCRIPTION/<br>NAMESPACE)                                                            | Consider<br>packaging<br>functions/<br>as an<br>installable<br>package<br>if long-term<br>mainte-<br>nance value<br>justifies it                               | CHAOS<br>Folder<br>structure                            |
| CS-006              | data/tidy/,<br>data/sensitive/<br>02_setup_and_global_variables.R:59 | Coding<br>standards              | Info     | Full repro-<br>ducibility<br>requires<br>code +<br>confidential<br>data +<br>pinned<br>packages;<br>only code<br>ships<br>publicly | Document<br>in the<br>README<br>(once<br>restored)<br>that a<br>data-use<br>agreement<br>or synthetic<br>data are<br>required for<br>outside re-<br>production | CHAOS<br>Repro-<br>ducibility                           |

| ID     | File:line                                     | Category         | Severity | Issue                                                                                                         | Recommended change                                                                                 | Checklist item(s)                        |
|--------|-----------------------------------------------|------------------|----------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------|
| CS-010 | scripts/R/analyzing_DSA.R:265,292,306,414,410 | Coding standards | Info     | in-loop pattern inconsistent with the pre-allocated-list pattern used elsewhere in the same file              | Replace growing-rbind loops with append-to-list + single do.call(rbind, ...) at the end            | TECH-VER Stage 5                         |
| CS-011 | scripts/R/functions_multi_model_cea.R:11      | Coding standards | Info     | Filename is a leftover from the pre-refactor multi-model architecture                                         | Rename to e.g. cea_helper or single_model_cea updating source() call sites                         | CHAOS Consistent style; TECH-VER Stage 5 |
| CS-013 | scripts/R/tests                               | Coding standards | Info     | No testthat usage anywhere; ad-hoc cat()-output diagnostic scripts; tests/tlr.R duplicates data-loading logic | Consider migrating highest-value scripts to test-that::test_that() with explicit assertions for CI | CHAOS Documentation, Modularity          |

## Appendix B — Function inventory & dependency map

Parsed 39 .R file(s) and 386 {r} chunk(s) from 51 Rmd/qmd file(s); found 207 user-defined function(s). Chunk-derived entries are tagged `file#chunk-label` with lines mapped to the original document. Code was parsed statically, never executed.

**Duplicate definitions** (same name defined more than once — last `source()` wins silently): `strategy_label`, `format_eur`, `format_icer`, `biomarker_label`, `restricted_mean`, `plot_dag`, `complete_data`, `fmt_num`, `fmt_p`, `fmt_hr`, `fmt_or`, `run_wilcox_test`, `run_fisher_test`, `run_mh_test`, `run_firth_cox_test`, `run_firth_logistic_test`, `run_permutation_test`, `edge_support_label`, `ci_support_label`, `predict_pop_avg`, `extract_km_data`, `fit_firth_cox`,

`extract_coxphf_results`, `get_interaction_hr`, `extract_ridge_interaction`, `get_strategies`, `get_biomarkers`, `fmt_eur`, `fit_all_direct` (see CS-007 for the risk this poses).

**Core call chain:** `get_current_model_config()` → `get_strategies()/get_biomarkers()/get_strategy_formula()` → `model_fun()` → `generate_psa_control_predictions()/generate_psa_population_averaged_predictions()` (PSA mode) or population-averaged prediction functions (deterministic mode) → `calculate_outcomes()`.  
`run_basecase()` (in `multi_model_cea.R`, despite its name — see CS-011/TR-CONS-007) orchestrates the deterministic base case; `run_psa_analysis()` orchestrates PSA; `run_evppi_analysis()/run_scenario_analysis()` sit downstream of the PSA cache.

**Key functions and locations** (selected): `model_fun` (`scripts/R/functions/model_fun.R:97`) — main PSM engine; `calculate_outcomes` (`scripts/R/functions/calculate_outcomes.R:2`) — cost/QALY aggregation, called by `model_fun` and `run_enriched_analysis`; `validate_model_params` (`model_fun.R:38`); `get_current_model_config/get_strategies/get_biomarkers/get_strategy_formula/get_strategy_formula()` (`model_configs.R:44-136`) — single source of truth for the strategy/biomarker/formula set; `generate_population_averaged_predictions / generate_population_averaged_predictions_multimodel` (`prediction_functions.R:39,208`); `run_basecase/load_psa_cache/create_ceac_plot/create_psa_summary` (`multi_model_cea.R:33-235`); `run_psa_analysis/generate_psa_samples` (`psa_functions.R:8,47`); `run_evppi_analysis/calculate_evppi_improved/calculate_population_evppi` (`evppi_functions.R:12,187`); `run_scenario_psa/run_scenario_evppi/run_all_scenarios` (`scenario_analysis.R:51,122,216`); `get_git_commit/collect_metadata/select_snapshots_for_comparison` (`snapshot_utils.R`) — snapshot/bug-fix-impact tooling.

**Input/parameters-object candidates:** `l_params_base` (`07_basecase_input_parameters.R:51`) — the central parameter object; `param_distributions/param_groups` (`08_sampling.R:452,474`) — PSA distribution specifications.

Full folder tree, package table, and the complete 207-row function inventory with per-function roxygen2 status, arguments, internal calls, and callers are preserved in `<SCRATCH>/inventory.md` and `<SCRATCH>/environment.md`, copied alongside the raw findings JSON in the `findings/` subdirectory next to this report.

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*Generated by ValidateHE v1 on 2026-07-12. Reviewer of record: AI system (Claude Code). Under AdViSHE C1, use Partial rather than Pass when the only external computerized-model review is this ValidateHE AI-assisted review; Pass requires reviewed materials documenting independent human modelling experts, their qualifications, and where their review results are reported. Related work (not used by this tool): assertHE, VIVID.*